

Heart Disease Machine Learning Pipeline Flowchart (OOP)

11-Step Modular OOP Architecture • Author: Sardar Ahmed | Alpatron Technologies

Step 1: class DataPreparation

Load dataset, clean 0s in [chol, trestbps], normalize strings -> Datapreparation/

Step 2: class ExploratoryDataAnalysis

Statistical summaries & 9 publication figures -> EDA/

Step 3: class DataTransformation

Imputation (Median/Mode), One-Hot/Binary Encoding, StandardScaler -> TransformedData/

Step 4: class FeatureSelector

Drop non-predictive ID & high-missing cols [ca, thal, slope] -> selected_features_data.csv

Step 5: class DataSplitter

Stratified 80/20 train/test partition -> X_train, X_test, y_train, y_test

Step 6: class ModelTrainer

Train baseline Logistic Regression, Random Forest, and XGBoost models -> Models/

Step 7: class ModelEvaluator

Evaluate test metrics: Accuracy, Precision, Recall, Weighted F1 on unseen test data

Step 8: class DataSampler

SMOTE oversampling on training data to balance minority severity classes -> X_train_sampled.csv

Step 9: class CrossValidator

Execute Stratified 5-Fold Cross-Validation across candidate models

Step 10: class HyperparameterOptimizer

GridSearchCV tuning for XGBoost hyperparameters -> Models/tuned_xgboost.pkl

Step 11: class ModelComparator

Benchmark all models, plot accuracy bar chart, export best_model.pkl -> ModelComparison/